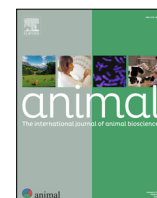




Animal

The international journal of animal biosciences



Enhancing the feeding value of corn stover for beef cattle via steam explosion: effects on rumen digestibility, growth and meat quality

S. Yi ^{a,b}, Y. Zhao ^a, H. Wu ^a, Q. Meng ^a, Z. Zhou ^{a,*}

^a State Key Laboratory of Animal Nutrition and Feeding, College of Animal Science and Technology, China Agricultural University, Beijing 100193, PR China

^b Frontier Technology Research Institute of China Agricultural University in Shenzhen, China Agricultural University, Shenzhen, Guangdong Province 518119, PR China



ARTICLE INFO

Article history:

Received 21 August 2025

Revised 26 February 2026

Accepted 2 March 2026

Available online 12 March 2026

Keywords:

Animal performance

Hydrothermal pretreatment

Lignocellulose

Physicochemical characteristics

Ruminant nutrition

ABSTRACT

To meet the rising global demand for premium animal protein, efficient utilisation of non-grain feedstuffs like corn stover has become critical for sustainable beef production. However, the high lignocellulosic content and structural rigidity of corn stover restrict its digestibility and nutritional availability in the rumen. This study assessed how steam explosion treatment influences the nutritional quality of corn stover and examined its effects on growth, serum biochemistry, slaughter characteristics, and meat quality of beef cattle. In Experiment 1, a rumen nylon bag study using three rumen-fistulated Simmental bulls showed that steam explosion significantly increased the *in situ* degradability of DM and organic matter, while reducing the fibre content. In Experiment 2, thirty Simmental bulls were randomly assigned to either a control group (CON) receiving 50% untreated corn stover in their roughage or a test group (SCS) where steam explosion-treated corn stover replaced the untreated portion. Over a 108-day trial, the SCS group exhibited significantly greater average daily gain (ADG) and dry matter intake (DMI) compared with CON ($P < 0.05$). Serum biochemical analysis revealed significantly elevated triglycerides, total protein, and high-density lipoprotein ($P < 0.05$), indicating enhanced protein metabolism and lipid transport. Slaughter performance was unaffected. However, meat from SCS-fed cattle showed higher pH and lower cooking and pressing losses ($P < 0.05$), indicating improved water-holding capacity. No differences were observed in meat nutrient composition. These results indicate that steam explosion serves as a valuable pretreatment for improving the ruminal digestibility and nutritional value of corn stover, thereby improving animal performance and meat quality in beef production systems, offering a promising strategy for utilising low-quality roughage in beef production.

© 2026 The Authors. Published by Elsevier B.V. on behalf of The animal Consortium. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Implications

Low digestibility of crop residues limits their use in sustainable beef production. This study investigated whether steam explosion could enhance the nutritional value of corn stover. The results demonstrate that this pretreatment improves ruminal degradability, resulting in increased weight gain and enhanced meat water-holding capacity in cattle. These findings suggest that steam explosion is a viable technology for converting low-quality agricultural by-products into high-quality feed. Implementing this method could reduce reliance on grain-based diets and improve production efficiency in the livestock industry, provided the process is economically optimised for large-scale application.

Introduction

In recent years, worldwide beef production and consumption have consistently risen to satisfy the increasing demand for premium animal protein (Greenwood, 2021). However, this trend places increasing pressure on the availability and cost of conventional feed resources (Pan et al., 2024). As a result, the strategic development and efficient utilisation of non-grain feedstuffs have become a critical focus for ensuring sustainable beef production systems.

Corn stover, the primary agricultural residue from maize cultivation, holds great promise as an alternative roughage resource for ruminants (Zhao et al., 2020). Despite its substantial global production, corn stover is often treated as waste in many regions—frequently burned in open fields, contributing to serious environmental pollution (Zhang et al., 2016). Only a limited proportion is currently utilised for biogas production (Fjortoft et al., 2019), soil amendment through field return (Li et al., 2018), or pro-

* Corresponding author.

E-mail address: zhouzm@cau.edu.cn (Z. Zhou).

cessed into feed (Fernandez et al., 2021). If effectively valorised, this abundant biomass could serve as a vast and renewable feed reservoir, playing an important role in the feed supply chain for ruminants.

Nevertheless, the broader application of corn stover as livestock feed is restricted by its high lignocellulosic content and poor digestibility (Loo et al., 2016). The presence of high levels of cellulose and hemicellulose, which form tightly bound complexes with lignin, limits the accessibility of fermentable carbohydrates to rumen microbes (Grabber, 2005). This structural rigidity significantly reduces its degradability and nutritional availability in the rumen. Therefore, to unlock the full nutritional potential of corn stover, it is essential to disrupt the lignin-carbohydrate or lignin-protein matrix, enhancing the availability of fermentable substrates.

The main pretreatment approaches for enhancing lignocellulose utilisation in biomass have involved physical, chemical, and biological techniques (Zhang et al., 2022). These methods work by modifying the fibre matrix structure, which increases the accessibility of carbohydrates and proteins to rumen microbial enzymes (Rothmann et al., 2023). Among these methods, steam explosion has gained increasing attention as a promising, eco-friendly pretreatment. It facilitates the cleavage of lignin-carbohydrate complexes through a combination of heat, pressure, auto-generated organic acids, and explosive decompression (Jacquet et al., 2015). Importantly, this autocatalytic process avoids the use of external chemical catalysts, which enhances its appeal as a potentially more sustainable strategy by circumventing issues of chemical cost, recovery, and effluent contamination, despite the inherent energy demand of the process (Chen et al., 2019). Substantial evidence from the bioenergy field has shown that steam explosion can induce significant physicochemical changes in lignocellulosic materials, enhancing enzymatic hydrolysis and sugar release (Nie et al., 2021; Zhang et al., 2023). In animal nutrition research, steam explosion has also been reported to reduce the levels of cellulose, hemicellulose, and lignin in straw-based feedstocks (Chang et al., 2012a, b), while improving *in vitro* digestibility and ruminal fermentation characteristics (He et al., 2022).

Although steam explosion has shown significant effects on the degradation of lignocellulose in the field of bioenergy, its application in ruminant production remains limited, especially in terms of systematic evaluation of growth, slaughter traits, and meat quality. To address this gap, this study proposes the following hypotheses: (1) Steam explosion treatment significantly alters the chemical composition of corn stover, thereby increasing the rate and extent of degradation; (2) Replacing traditional corn stover with steam-exploded corn stover in beef cattle diets can improve growth, serum metabolic parameters, slaughter traits and meat quality.

Material and methods

Steam explosion processing and sampling

Corn stover was harvested at the late milk stage, air-dried naturally to a moisture content of approximately 12%, and then cut into 3–5 cm pieces and stored in a well-ventilated and dry place. The chopped stover was then subjected to steam explosion pretreatment. The treatment was performed using industrial-scale equipment at a feed processing plant (Hefei Zhougongcao Co., Ltd, Anhui Province). The steam explosion conditions were as follows: temperature 200 °C, pressure 2 MPa, maintained for 5 min. The treated samples were then dried in an oven at 105 °C until the moisture content was below 15% for storage and subsequent experimental use.

Experimental design and animal management

This study included two experiments: experiment 1 was a rumen nylon bag experiment, and experiment 2 was a feeding and slaughtering experiment.

Experiment 1

Three rumen-fistulated Simmental bulls (450 ± 21 kg) were used to evaluate the ruminal degradability of corn stover. The trial was carried out at the Beijing Beef Cattle Research and Demonstration Base of China Agricultural University. The experimental diet was prepared in accordance with NASEM (2016) (Table 1). The animals were offered feed twice daily at 0800 and 1600 h and had unrestricted access to fresh water. Nylon bags (12 cm × 10 cm; 50 µm pore size) were cleaned and dried prior to use. Both steam-exploded and untreated corn stalks were dried at 105 °C and then ground using a hammer mill (SFSP66*80b, Jiangsu Degao Machinery Co., Ltd.), which is commonly used for processing agricultural samples. The dried samples were ground until they could pass through a 2 mm sieve. The ground samples were used for rumen in-situ degradation experiments. Approximately 3.5 g of each sample (steam-exploded or untreated corn stover) was weighed into a nylon bag, with 3 replicates per sample for each time point. The prepared nylon bags were averagely fixed on the Bang-Bang groove developed by the Beef Cattle Research Center of China Agricultural University (Wang et al., 2016), and then put into the rumen abdominal sac, where the test procedure adopted the principle of “take out and put in” with two time points each time. The nylon bags for time points 6 to 96 h (6, 12, 24, 48, 72 and 96 h) were placed into the rumen and removed at their respective set times and rinsed immediately with water, then dried at 65 °C for 48 h. For the 0 h time point, bags were not incubated in the rumen but were directly rinsed with water to establish the

Table 1
Ingredients and chemical composition of diets for fistulated Simmental bulls in Experiment 1.

Items (DM basis) ¹	Content
Ingredient composition	
Whole corn silage, %	40.00
Corn stover, %	10.00
Dry-pressed corn, %	24.00
Soya bean meal, %	10.00
Brewer's spent grain, %	10.00
Cottonseed meal, %	3.00
Salt, %	0.75
Calcium perphosphate, %	0.64
Calcium oxide, %	0.30
Sodium bicarbonate, %	0.30
Premix, %	1.00
Nutrient level	
CP, %	12.37
NDF, %	46.30
ADF, %	24.55
Ca, %	0.57
P, %	0.39
ME, Mcal/kg	2.62
Nem, Mcal/kg	1.72
Neg, Mcal/kg	1.11

¹ Each kilogram of premix contains the following: iron, 2.4 g; copper, 4 g; zinc, 2.6 g; manganese, 1.8 g; iodine, 0.16 g; selenium, 0.03 g; cobalt, 0.03 g; V_A, 6 800 IU; V_D, 800 IU; V_E, 50 IU. ME = metabolisable energy; NEm = net energy for maintenance; NEg = net energy for gain. ME, NEm, and NEg are calculated values using the NASEM (2016) model, while the remaining nutrient levels are directly measured.

baseline residue weight. The amount of material washed out during this initial rinse (0 h) was included in the degradation calculation, as this material was considered part of the degradable fraction. The residual amount of the 0-hour sample was set as the baseline for calculating the degree of degradation at subsequent time points. The sample residues were ground through a 1 mm sieve and packed in ziplock bags for the measurement of chemical components.

Experiment 2

Thirty Simmental bulls (397 ± 57 kg) were randomly assigned to two treatments: a control (CON) group and a steam-exploded corn stover (SCS) group, each comprising 15 animals. The trial lasted 108 days in total, consisting of an 18-day adaptation phase followed by a 90-day feeding phase. The experimental diet was prepared according to NASEM (2016) (Table 2). The control diet (CON) contained 20% corn stover, which was replaced by steam-exploded corn stover in the treatment diet (SCS). All animals were housed in individual tie stalls and provided with wheat straw as bedding material to ensure comfort. They received feed twice daily at 0400 and 1600 h. Two experimental diets were gradually introduced during the adaptation period to allow the animals to adapt and reduce stress caused by dietary changes. Initial BW was measured at the start of the adaptation period. Throughout the experiment, they had unrestricted access to fresh drinking water. Feed samples were collected once a week for DM determination to ensure the accuracy of the feeding amount. After the 108-day feeding trial, eight bulls per treatment were randomly selected and transported to a commercial slaughterhouse (located 20 km from the feedlot, approximately 1 h of transport). To minimise stress and ensure accurate live weight measurement, the bulls were fasted for 12 h with free access to water prior to slaughter. The cattle were slaughtered following standard humane procedures, including mechanical captive bolt stunning followed by immediate

exsanguination. All procedures were conducted in compliance with the relevant national guidelines for the care and use of agricultural animals.

Sample collection

Corn stover samples were randomly taken both prior to and following the steam explosion treatment. The collected samples were air-dried at 65 °C and then ground to pass through a 1 mm sieve for subsequent chemical composition analysis. The daily feed offered, and refusals for each animal were recorded to determine DM intake. Bulls were weighed prior to the morning feeding on both the first and final days of the trial to calculate average daily gain. On the last day, before feeding, blood samples were collected via jugular venipuncture and centrifuged at $1\,200 \times g$ for 10 min to obtain serum. Serum biochemical parameters were measured using an automated analyzer (Hitachi 7600, Japan) with commercial kits provided by the Nanjing Jiancheng Bioengineering Institute (Nanjing, China). Specifically, glucose (GLU, Cat. No. F006-1-1), total cholesterol (TC, Cat. No. A111-1-1), and triglycerides (TG, Cat. No. A110-1-1) were determined via the GPO-PAP enzymatic method. Total protein (TP, Cat. No. A045-4-1) was measured using the Coomassie brilliant blue method, while albumin (ALB, Cat. No. A028-1-1) was determined by the bromocresol green method. Blood urea nitrogen (BUN, Cat. No. C013-2-1) was measured using the urease-glutamate dehydrogenase method. High-density lipoprotein cholesterol (HDL-C, Cat. No. A112-1-1) and low-density lipoprotein cholesterol (LDL-C, Cat. No. A113-1-1) were determined by the direct surfactant clearance method. Very low-density lipoprotein cholesterol was measured using a selective precipitation method (Cat. No. A114-1-1). Serum globulin was calculated as the difference between TP and ALB. All assays were performed strictly following the manufacturer's protocols to ensure high specificity and sensitivity.

Bulls were weighed prior to slaughter, and carcass weights were documented postslaughter. All animals were fasted for 12 h before slaughter, and the slaughter weight was the weight after fasting, which was different from the final weight at the end of the experiment. Following a 48-hour ageing period, bone and boneless meat weights were measured for each animal. Samples from the area between the 5th and 6th ribs were collected to assess meat quality and nutrient composition. The eye muscle area was determined by sawing the spine at the rear edge of the 5th rib of the carcass, cutting the bottom 5–6 ribs, tracing the eye muscle area with sulfuric acid paper at the rear edge of the 5th rib, and calculating the area with a planimeter. The back fat thickness was measured with a steel ruler at a position about 3 to 5 cm from the midline of the back between the 5th and 6th ribs.

Chemical analysis and calculation

Feed and meat samples were analysed for dry matter (DM; method 934.01) and ash (method 942.05) following AOAC (2007) procedures. Nitrogen content, including neutral detergent insoluble nitrogen and acid detergent insoluble nitrogen, was measured using the Dumas combustion technique (RaPip N III, Elementar Analysensysteme, Germany), and CP levels were calculated as nitrogen $\times 6.25$. Ether extract was determined with an automated extraction unit (ANKOM XT101, ANKOM Technology, Macedon, NY, USA). NDF, ADF, and acid detergent lignin concentrations were analysed using a fiber analyzer (ANKOM A220, ANKOM Technology, Macedon, NY, USA) according to the methods of Van Soest et al. (1991) and Van Soest and Robertson (1981).

The rumen degradation extent was calculated using the model established by Ørskov and McDonald (1979):

Table 2
Ingredients and chemical composition of diets for Simmental bulls in Experiment 2.

Items (DM basis) ¹	Treatment ²	
	CON	SCS
Ingredient composition		
Whole corn silage, %	20.00	20.00
Corn stover, %	20.00	0.00
Steam-exploded corn stover, %	0.00	20.00
Crush corn, %	39.00	39.00
Soybean meal, %	4.80	4.80
Cottonseed meal, %	4.80	4.80
Rapeseed meal, %	2.40	2.40
Corn germ meal, %	2.40	2.40
Molasses, %	1.20	1.20
Distillers dried grains with solubles, %	3.00	3.00
Sodium chloride, %	0.60	0.60
Sodium bicarbonate, %	0.60	0.60
Premix, %	1.20	1.20
Chemical composition		
CP, %	13.14	13.05
NDF, %	34.91	29.74
ADF, %	19.50	21.22
Ca, %	0.55	0.57
P, %	0.37	0.40
ME/(Mcal/kg)	2.78	2.87
NEm/(Mcal/kg)	1.86	1.88
NEg/(Mcal/kg)	1.24	1.28

¹ Each kilogram of premix contains the following: iron, 2.4 g; copper, 4 g; zinc, 2.6 g; manganese, 1.8 g; iodine, 0.16 g; selenium, 0.03 g; cobalt, 0.03 g; V_A, 6 800 IU; V_D, 600 IU; V_E, 50 IU. ME = metabolisable energy; NEm = net energy for maintenance; NEg = net energy for gain. ME, NEm, and NEg are calculated values using the NASEM (2016) model, while the remaining nutrient levels are directly measured.

² CON = control group; SCS = steam-exploded corn stover group.

$$dp = a + b(1 - e^{-ct})$$

where dp is the extent of degradation at time t , a (%) is the rapidly degradable fraction, b (%) is the slowly degradable fraction, c (%/h) is the degradation rate constant of b , and t is the incubation time of the sample in the rumen.

The rumen effective degradation proportion was calculated with the following formula:

$$ED = a + b \times c / (c + k)$$

where k is the outflow rate of rumen, calculated by $k = 0.0139 + 0.1775c$.

These equations are based on the model developed by Ørskov and McDonald (1979) to estimate the degradation of feed components in the rumen.

Meat quality

Following the ageing period, the pH of the longissimus lumborum muscle was determined at 25 °C using a calibrated pH meter (MATTHAUS, Germany), with three readings taken per sample and averaged. After a 1 h blooming period, muscle colour parameters, including lightness (L^*), redness (a^*), and yellowness (b^*), were evaluated using a CR400/410 meat colorimeter (Minolta, Osaka, Japan). The instrument was standardised against a white calibration plate prior to use, and measurements were performed under D65 illumination with a CIE 10° observer and an 8 mm aperture. Each sample was assessed in triplicate, and the mean values were used for analysis (Yi et al., 2024b).

Cooking traits assessed were pressing loss, cooking loss, and shear force. To avoid variation among cooking batches, all samples were processed in a single batch. Measurements were performed following the procedure described by He et al. (2018), with the steps summarised below.

A 20 mm × 20 mm × 20 mm meat block was taken, weighed and placed between two layers of gauze. Eighteen layers of qualitative medium-speed filter paper on the top and bottom and a hard plastic plate were padded outside the filter paper. The assembly was placed on a texture analyser (Food Technology Corporation, ILC-S 1000N). A pressure of 35 kg was applied for 5 min. Subsequently, the meat block was removed, weighed again, and the pressing loss rate was calculated.

A meat portion of uniform size (60 mm × 60 mm × 40 mm) was trimmed of surface fascia and fat, sealed in a plastic bag, and cooked in an 80 °C water bath until its core temperature reached 70 °C. Each longissimus muscle sample was analysed in triplicate, and the mean value was recorded. Cooking loss was determined by calculating the difference between the pre- and postcooking weights and expressing it relative to the initial weight.

Shear forces were measured cross-sectionally using a texture analyser (TMS-PRO, FTC, USA) by drilling and cutting holes parallel to the muscle fibre direction in cooked meat using a circular sampler; four measurements were performed for each long-distance muscle sample, and the average was calculated.

Statistical analysis

All data were analysed using SPSS software (version 21.0, IBM Corp., Armonk, NY, USA) and are expressed as mean ± SE of the mean (SEM). In Experiment 1 (rumen nylon bag study), the individual rumen-fistulated bull ($n = 3$) was considered the experimental unit. For each bull, values from replicate nylon bags were averaged before statistical analysis. The data were analysed using a mixed model with treatment as a fixed effect and animal as a random effect, according to the following model:

$$Y_{ij} = \mu + T_i + A_j + e_{ij}$$

where Y_{ij} is the observation, μ is the overall mean, T_i is the fixed effect of treatment (steam-exploded vs. untreated corn stover), A_j is the random effect of animal ($j = 1-3$), and e_{ij} is the residual error.

In Experiment 2 (feeding and slaughter trial), the individual bull ($n = 15$ per group) was the experimental unit. Performance, serum metabolites, carcass traits, and meat quality data were analysed using independent-sample t -tests to compare the CON and SCS groups. Significance was declared at $P < 0.05$, with $0.05 \leq P < 0.10$ considered a tendency.

Results

Effect of steam explosion treatment on chemical compositions of corn stover

As presented in Table 3, steam explosion elevated the DM, CP, neutral detergent insoluble nitrogen, acid detergent insoluble nitrogen, ether extract, and ash levels of corn stover ($P < 0.05$) while decreasing NDF and ADF contents ($P < 0.05$). Acid detergent lignin levels, however, were not significantly affected.

Effect of steam explosion treatment on in situ ruminal degradation of corn stover

Figs. 1 and 2 depict the effects of steam explosion treatment on rumen degradation kinetics and model parameters for DM, OM, NDF, and ADF. Steam explosion significantly increased DM and OM degradation at all-time points ($P < 0.01$; Fig. 1). For NDF and ADF, degradation dropped significantly at 0 and 6 h ($P < 0.01$) but was markedly higher at 48, 72, and 96 h ($P < 0.01$). No significant differences were detected at 12 or 24 h.

Steam explosion significantly elevated the a , b , ($a+b$), and ED values for DM and OM ($P < 0.01$; Fig. 2). A tendency towards reduced c -value was observed for DM ($P = 0.082$), while OM model parameters (c , k) remained unaffected. For NDF and ADF, the treatment significantly decreased the a -value ($P < 0.05$) but increased b and ($a+b$) ($P < 0.01$). Parameters c , k , and ED showed no significant changes.

Effect of replacement of corn stover by steam-exploded corn stover on growth and serum biochemical indices in beef cattle

Replacing conventional corn stover with SCS resulted in significantly higher ADG and DMI. However, no significant differences

Table 3

Effects of steam explosion treatment on the nutritional chemical composition of corn stover fed to Simmental bulls.

Item (DM basis) ¹	Treatment ²		SEM	P-value
	CON	SCS		
DM, %	92.8	94.5	0.08	<0.01
CP, %	4.15	5.38	0.33	<0.01
NDF, %	69.1	51.6	0.41	<0.01
ADF, %	44.8	43.5	0.42	0.01
ADL, %	7.06	7.93	0.45	0.13
NDIN, %	0.25	0.46	0.03	<0.01
ADIN, %	0.25	0.50	0.06	<0.01
EE, %	1.66	2.40	0.31	<0.01
Ash, %	5.44	8.95	0.16	<0.01

¹ NDIN = neutral detergent insoluble nitrogen; ADIN = acid detergent insoluble nitrogen; EE = ether extract.

² CON = control group; SCS = steam-exploded corn stover group. Data represent averages of three rumen-fistulated bulls ($n = 3$); analytical replicates within animal were averaged before statistical analysis.

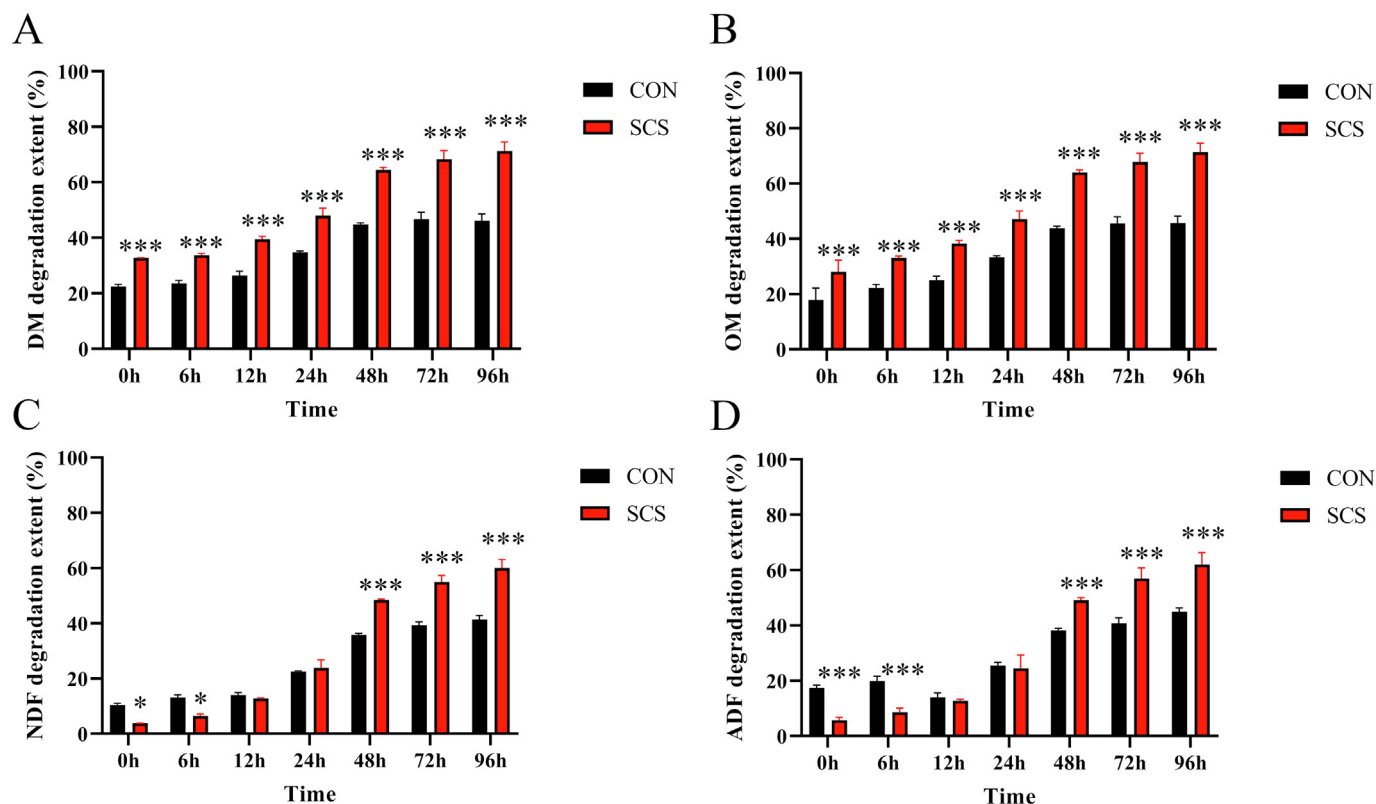


Fig. 1. Effect of steam explosion on the degradation of DM (A), OM (B), NDF (C), and ADF (D) of corn stover in the rumen of Simmental fistulated bulls. * 0.01 < P < 0.05, *** P < 0.001, n = 3 rumen-fistulated bulls. Replicate nylon bags per sample were averaged within animal before analysis. OM = organic matter; CON = control group (untreated corn stover); SCS = steam-exploded corn stover group.

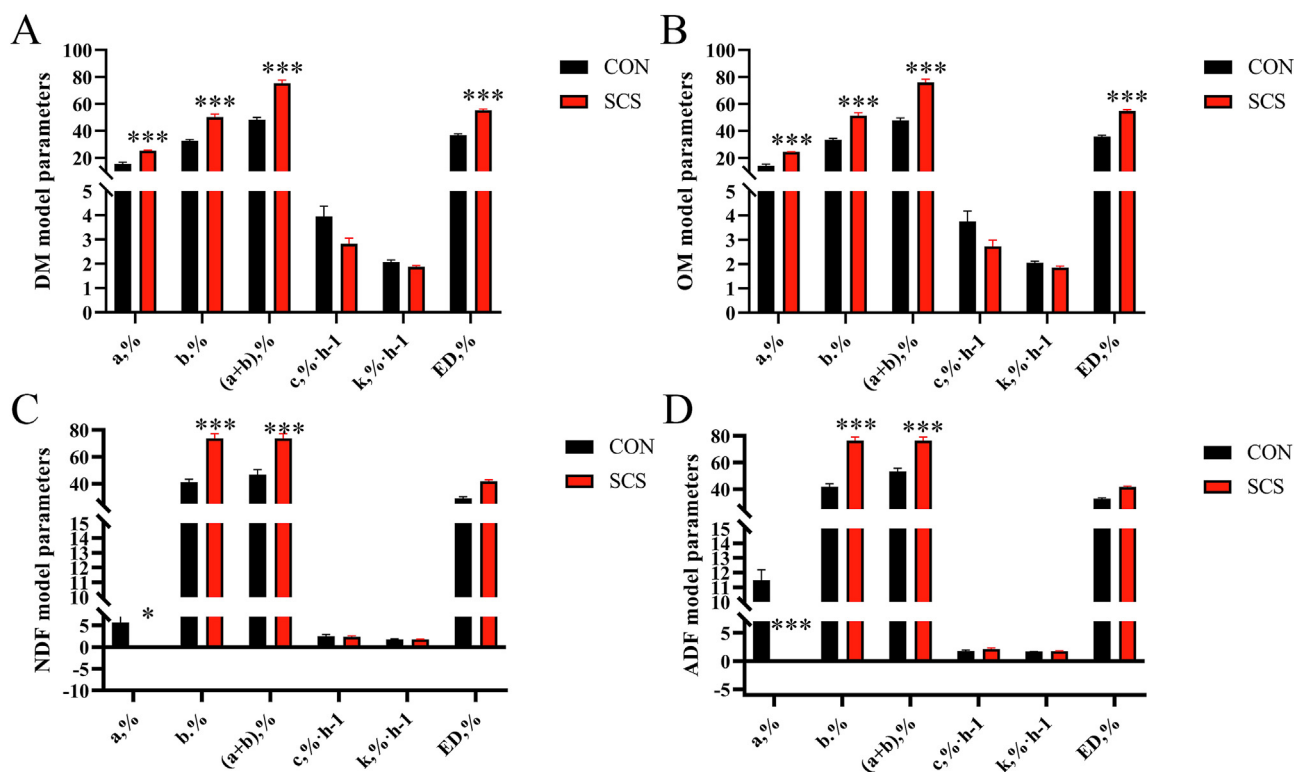


Fig. 2. Effects of steam explosion on rumen dynamic model parameters of DM (A), OM (B), NDF (C) and ADF (D) of corn stover in the rumen of Simmental fistulated bulls. In the model, parameter 'a' is constrained to be non-negative. * 0.01 < P < 0.05, *** P < 0.001, n = 3 rumen-fistulated bulls. Replicate nylon bags per sample were averaged within animal before analysis. OM = organic matter; CON = control group (untreated corn stover); SCS = steam-exploded corn stover group; a = rapidly degradable fraction; b = slowly degradable fraction; c = constant rate of degradation of fraction b; k = rumen outflow rate; ED = effective degradability.

were observed in initial BW, final BW, or ADG/DMI between the two groups (Table 4).

Serum biochemistry results (Table 5) showed significantly elevated TG, TP and HDL-C concentrations in the SCS group ($P < 0.05$). In addition, globulin, BUN and very low-density lipoprotein cholesterol levels exhibited an upward trend in cattle fed SCS.

Effect of replacement of corn stover by steam-exploded corn stover on carcass characteristics and meat quality in beef cattle

Carcass characteristics did not differ significantly between groups (Table 6). For meat quality (Table 7), SCS-fed cattle exhib-

Table 4

Effects of steam-exploded corn stover as a substitute for corn stover on growth performance of beef cattle.

Item ¹	Treatment ²		SEM	P-value
	CON	SCS		
IBW, kg	399	396	21.18	0.89
FBW, kg	549	561	23.63	0.60
ADG, kg	1.39	1.53	0.05	0.01
DMI, kg/d	9.83	11.35	0.30	<0.01
ADG/DMI	0.14	0.14	0.004	0.13

¹ IBW = initial BW; FBW = final BW; ADG = average daily gain; DMI = DM intake.

² CON = control group; SCS = steam-exploded corn stover group. n = 15 bulls per group. The individual bull was the experimental unit.

Table 5

Effects of steam-exploded corn stover as a substitute for corn stover on serum biochemical parameters of beef cattle.

Item ¹	Treatment ²		SEM	P-value
	CON	SCS		
GLU (mmol/L)	5.69	5.76	0.27	0.80
TC (mmol/L)	2.73	3.00	0.18	0.15
TG (mmol/L)	0.34	0.38	0.02	0.05
TP (g/L)	62.2	64.6	1.07	0.03
ALB (g/L)	27.9	28.3	0.37	0.33
GLB (g/L)	34.3	36.3	1.07	0.07
BUN (mmol/L)	3.91	4.37	0.23	0.06
LDL-C (mmol/L)	1.48	1.66	0.14	0.21
HDL-C (mmol/L)	0.68	0.90	0.09	0.02
VLDL-C (mmol/L)	0.26	0.28	0.01	0.06

¹ GLU = glucose; TC = total cholesterol; TG = triglycerides; TP = total protein; ALB = albumin; GLB = globulin; BUN = blood urea nitrogen; LDL-C = low-density lipoprotein cholesterol; HDL-C = high-density lipoprotein cholesterol; VLDL-C = very low-density lipoprotein cholesterol.

² CON = control group; SCS = steam-exploded corn stover group. n = 15 bulls per group. The individual bull was the experimental unit.

Table 6

Effect of steam-exploded corn stover as a substitute for corn stover on carcass characteristics of beef cattle.

Item	Treatment ¹		SEM	P-value
	CON	SCS		
Slaughter weight, kg	615	636	15.39	0.20
Carcass weight, kg	338	347	9.99	0.39
Dressing percentage, %	54.9	54.5	1.21	0.78
Bone weight, kg	45.3	46.2	1.64	0.62
Net meat weight, kg	293	301	9.40	0.41
5th rib fat thickness, cm	1.50	1.42	0.25	0.78
Longissimus lumborum muscle area, cm ²	38.9	39.5	2.08	0.80
Net meat percentage, %	47.6	47.3	1.11	0.82
Net meat weight/bone weight	6.46	6.51	0.27	0.90

¹ CON = control group; SCS = steam-exploded corn stover group. n = 8 bulls per group. The individual bull was the experimental unit.

ited significantly higher pH ($P < 0.05$) and lower pressing loss and cooking loss ($P < 0.05$) compared to CON. A decreasing trend was observed for L^* , a^* , and shear force in SCS beef. Nutrient content of beef did not differ between treatments.

Discussion

Developing non-grain feed resources and broadening feed supply channels can effectively alleviate the competitive pressure between grain and feed (Getahun et al., 2025). This study provides evidence that steam explosion significantly changes the chemical composition of corn stover, with a shift of cell wall components from detergent-insoluble to soluble pools and a resultant increase in the concentration of CP, while reducing fibre content. These changes collectively improve its rumen degradation rate. Consistent with our first hypothesis, steam-exploded corn stover showed improved *in situ* degradation of DM and organic matter, and the fibre degradation kinetics changed favourably during the extended incubation period. Furthermore, replacing untreated corn stover with steam-exploded stover significantly increased DMI and ADG, without adversely affecting FBW, slaughter traits, or muscle nutritional composition. On the contrary, meat quality attributes were improved, including increased muscle pH and reduced cooking and pressing losses, which indicated enhanced water retention capacity, which supported our second hypothesis.

A key finding is the significant reduction in NDF and ADF content of corn stover after steam explosion. This reduction is mainly attributed to the partial dissolution and structural destruction of hemicellulose during the high-pressure cooking and decompression stages of steam explosion (Jacquet et al., 2015; Zhang et al., 2023). While hemicellulose is an important component of NDF, its contribution to the ADF fraction is minimal. The substantially greater reduction in NDF compared to ADF is consistent with the selective degradation of hemicellulose by steam explosion, a component not reflected in the ADF value. Elevated temperature and pressure disrupt the hydrogen and ester bonds of the lignocellulosic structure, hydrolyse hemicellulose-bound acetyl groups to release organic acids (such as acetic acid), and subsequently promote the breakdown of hemicellulose and amorphous cellulose (Chang et al., 2012a, b; Nie et al., 2021). Notably, acid detergent lignin levels showed no significant change, a finding consistent with earlier reports (He et al., 2022). This suggests that steam explosion primarily acts by altering the association between lignin and the more digestible components of the plant cell wall, such as hemicellulose and cellulose, thereby making them more accessible for microbial degradation, rather than large-scale degradation of lignin polymers themselves (Grabber, 2005). Studies have reported that shear forces during decompression physically disrupt the fibre structure and increase the surface area, but do not necessarily remove the core lignin fraction that can be measured as acid deter-

Table 7

Effect of steam-exploded corn stover as a substitute for corn stover on beef quality.

Item ¹	Treatment ²		SEM	P-value
	CON	SCS		
Quality trait				
pH	5.48	5.58	0.04	0.03
<i>L</i> [*]	36.8	35.1	0.82	0.07
<i>a</i> [*]	17.9	19.6	0.83	0.07
<i>b</i> [*]	8.19	8.05	0.69	0.14
Pressing loss, %	23.3	18.9	1.88	0.04
Cooking loss, %	37.8	34.3	1.47	0.04
Shear force, N	75.7	64.9	5.80	0.08
Nutrient content				
Moisture, %	74.8	75.4	0.34	0.74
CP, %	83.2	84.5	0.64	0.33
EE, %	11.6	10.7	0.77	0.58
Ash, %	4.35	4.44	0.18	0.60

¹ *L*^{*} = lightness; *a*^{*} = redness; *b*^{*} = yellowness; EE = ether extract.² CON = control group; SCS = steam-exploded corn stover group. n = 8 bulls per group. The individual bull was the experimental unit.

gent lignin (Li et al., 2024). The significant increases in DM, CP, neutral detergent insoluble nitrogen, acid detergent insoluble nitrogen, ether extract, and Ash are notable. The apparent increase in the concentration of components such as CP, ether extract, and ash on a DM basis is attributed to the loss of volatile organic compounds that constitute part of the DM itself during the high-temperature steam explosion and subsequent drying process. These volatiles may include organic acids (e.g., acetic acid), furan derivatives (e.g., furfural), and other degradation products released from hemicellulose and amorphous cellulose (Jacquet et al., 2015; Zhang et al., 2023). This loss, coupled with the substantial degradation and removal of structural carbohydrates (NDF/ADF), concentrates the remaining non-volatile components in the treated residue. The apparent increase in CP concentration is mainly attributed to the relative effect of the concentration of remaining nitrogenous compounds, which resulted from the loss of structural carbohydrates (NDF and ADF) during the steam explosion treatment. While it is likely that some of the hemicellulose and cellulose fractions were reduced in size and became soluble in neutral detergent solution, we did not directly measure the solubility of these compounds. The increase in CP concentration, therefore, can be attributed to the concentration of nitrogenous compounds resulting from the substantial loss of low-nitrogen or nitrogen-free components, primarily structural carbohydrates (hemicellulose and amorphous cellulose) and their volatile degradation products (e.g., organic acids, furans), during processing and drying (Chang et al., 2012a, b; Jacquet et al., 2015). However, the large increase in neutral detergent insoluble nitrogen and acid detergent insoluble nitrogen indicates that a considerable part of the protein is bound or complexed to the lignocellulosic matrix during steam explosion, which may be achieved through the Maillard reaction of reducing sugars (released from hemicellulose/cellulose) with amino acids/proteins under high temperature and pressure (Rothmann et al., 2023). This forms acid detergent-insoluble complexes, which reduce the true utilisation of proteins despite the higher CP values. The increase in ether extract may be due to the concentration effect and/or the release of lipid components previously entrapped in the lignocellulosic structure. While the standard ether extraction procedure is widely used for lipid recovery from plant cell walls, the possibility that some lipid components may remain trapped within the cell wall matrix and not be fully recovered by this method remains a subject for future investigation. The increase in ash content on a DM basis is primarily attributed to the concentration of minerals resulting from the substantial loss of organic matter during steam explosion and subsequent drying. This loss includes volatile organic compounds (e.g.,

organic acids, furans) and, predominantly, the degradation and removal of structural carbohydrates (NDF/ADF). These compositional changes highlight that steam explosion fundamentally reshapes the structure of the stover and reduces structural constraints.

Steam explosion enhanced the ruminal degradation of corn stover. The physical destruction and chemical decomposition of hemicellulose and amorphous cellulose made a larger proportion of the substrate immediately available (component a) and more easily colonised/fermented by rumen microorganisms (component b and higher ED) (Shi et al., 2019; Zhao et al., 2018). The rate constant (c) of DM showed a decreasing trend, indicating that steam explosion has likely shifted some of the potentially degradable fraction 'b' to the immediately available fraction 'a', leaving more slowly degradable components in the remaining fraction 'b'. This shift may be due to the formation of some refractory products or changes in lignin distribution, but it does not offset the significant improvement in ED overall (He et al., 2022; Zhao et al., 2018).

The degradation kinetics of NDF and ADF showed more complex dynamics. After 6 h, the extent of degradation was lower for steam-exploded than for untreated corn stover, in contrast to the improvements in DM and OM degradation. This likely reflects the initial flushing of highly soluble non-fibre components from the steam-exploded stover bags, such as dissolved hemicellulose fragments, sugars, and organic acids. Since NDF/ADF represents the fibre residue after extraction, the removal of these soluble components leaves behind a fibre residue that, after 6 h, appears less degradable than untreated stover, which initially contains less soluble material. However, this pattern shifted at later time points (48, 72, and 96 h). The physical disruption and chemical decomposition induced by steam explosion create a fibre structure that, although initially appearing inaccessible (low a value), becomes more susceptible to microbial enzymes over time (Du et al., 2019; Wang et al., 2025). This was further supported by the significant increase in the potentially degradable fraction (b), total extent of degradation (a+b), and ED of NDF and ADF. The rate constant for fibre degradation (c) remained unchanged, indicating that microorganisms degraded the potentially degradable fibre fraction (b) at a similar rate after colonisation, but the amount of fibre they were able to degrade (b and ED) was significantly higher in steam-exploded corn stover. This delayed but superior fibre degradation is a key advantage, promoting continued fermentation in the rumen and potentially improving energy availability for a longer period (Zhao et al., 2018). It emphasises that short-term (< 24 h) degradation studies may underestimate the true benefits of steam explosion on fibre utilisation.

Steam explosion significantly increased ADG and DMI in the SCS group, indicating that replacing untreated stover with steam-exploded stover enhanced cattle growth rate and feed intake. However, the increase in ADG was not associated with improved feed conversion efficiency under the conditions of this experiment. Essentially, steam explosion allowed cattle to consume more feed and digest it faster, which resulted in faster growth, but did not translate into higher feed efficiency. This increase in voluntary feed intake strongly suggests that steam-exploded stover is more easily fermented in the rumen, likely due to a reduction in rumen fill, rather than improved palatability (Yang et al., 2025). Additionally, the reduced physical structure (lower NDF/ADF content, smaller particle size) and potential changes in texture may enhance the acceptability of SCS diets (Fernandez et al., 2021; Hussein et al., 2024). The significant increase in ADG with more DMI was also echoed by the changes in blood parameters. Increased TP suggests enhanced protein metabolism, which may result from increased microbial protein synthesis (higher degradation rate of OM in the rumen) or increased amino acid absorption (Yi et al., 2025). Increased Glu content and trends in BUN, Very low-density lipoprotein cholesterol and TG content suggest increased energy absorption and enhanced lipid metabolism, likely due to increased absorption of fatty acids as a result of higher DMI, which is reflected in the elevated concentrations of Very low-density lipoprotein cholesterol and TG in blood (Yi et al., 2024a). Significantly increased HDL-C content, which is often associated with reverse cholesterol transport, may reflect enhanced lipid metabolism, specifically the more active transfer of cholesterol from peripheral tissues to the liver (Rosenson et al., 2012).

Carcass characteristics, including slaughter weight, carcass weight, and fat deposition, did not differ significantly between the CON and SCS groups. The absence of negative effects on carcass traits is noteworthy, as it demonstrates that the significant improvements in growth performance (increased ADG and DMI) and meat quality (improved water-holding capacity) achieved with SCS supplementation were not attained at the expense of carcass value.

Replacing conventional corn stover with SCS in beef cattle diets improved key meat quality traits, including muscle pH, water holding capacity (as measured by pressing loss and cooking loss), and colour. The longissimus dorsi muscle pH was significantly higher in the SCS group compared to the CON group. This moderate increase in postmortem pH is typically indicative of improved glycogen retention before slaughter, which helps minimise protein denaturation and enhance muscle water holding capacity (Li et al., 2025; Yu et al., 2025). Consistent with the pH results, the pressing loss and cooking loss in the SCS group were significantly reduced, indicating an increase in water holding capacity. This improvement has important practical significance because lower pressing loss during processing not only maintains carcass weight but also improves juiciness, which is a key sensory attribute of meat (Yang et al., 2022). The improvement in water holding capacity may be attributed to the more stable muscle protein structure under more favourable pH conditions, thereby preventing excessive exudate loss (Zhang et al., 2021). In terms of colour parameters, the SCS group showed a trend of reduced brightness and increased redness. Red meat is generally preferred by consumers, often indicating a higher myoglobin concentration or slower myoglobin oxidation (Xu et al., 2022). The limited impact of SCS on tenderness and meat chemical composition aligns with its primary effect of enhancing growth performance without altering carcass characteristics. While SCS improved water-holding capacity, a key quality trait, the absence of significant effects on shear force and nutrient profile suggests that the substitution did not substantially alter

the metabolic pathways governing these specific attributes under the conditions of this study.

Conclusions

This study demonstrated that steam explosion improves the nutritional value of corn stover for growing beef cattle. Replacing untreated corn stover with SCS significantly increased DMI and ADG, reflecting enhanced nutrient absorption and metabolism. SCS feeding also improved key meat quality traits, including muscle pH, water holding capacity, and cooking loss. These findings highlight SCS as a promising and sustainable feed ingredient that can enhance beef production efficiency and meat quality.

Ethics approval

All animal care and experimental procedures were reviewed and approved by the Animal Welfare Committee of China Agricultural University (approval number: AW81404202-1-7).

Data and model availability statement

The data/models were not deposited in an official repository. Data and model will be made available on request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) did not use any AI and AI-assisted technologies.

Author ORCIDs

Simeng Yi: <https://orcid.org/0000-0001-7616-427X>.

Yongfeng Zhao: <https://orcid.org/0009-0006-5085-9931>.

Hao Wu: <https://orcid.org/0000-0003-1320-4603>.

Qingxiang Meng: <https://orcid.org/0000-0002-7093-9027>.

Zhenming Zhou: <https://orcid.org/0000-0001-8780-7170>.

CRediT authorship contribution statement

S. Yi: Writing – review & editing, Writing – original draft, Visualisation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualisation. **Y. Zhao:** Visualisation, Methodology, Data curation. **H. Wu:** Writing – review & editing, Validation, Supervision, Resources, Conceptualisation. **Q. Meng:** Writing – review & editing, Validation, Resources, Conceptualisation. **Z. Zhou:** Writing – review & editing, Validation, Supervision, Resources, Conceptualisation.

Declaration of interest

The authors declare no conflicts of interest.

Acknowledgements

We thank all the staff of the Beef Cattle Research Center of China Agricultural University for their help in this study.

Financial support statement

This work was supported by the National Key R&D Program of China (2024YFD1301100, 2023YFD1301302) and the Government Purchase Service (grant numbers: 16200158, 16190050).

References

- Association of Official Analytical Chemists (AOAC), 2007. *Official Methods of Analysis of AOAC International*. AOAC Int., Gaithersburg, MD, USA.
- Chang, J., Cheng, W., Yin, Q., Zuo, R., Song, A., Zheng, Q., Wang, P., Wang, X., Liu, J., 2012a. Effect of steam explosion and microbial fermentation on cellulose and lignin degradation of corn stover. *Bioresource Technology* 104, 587–592. <https://doi.org/10.1016/j.biortech.2011.10.070>.
- Chang, J., Yin, Q.Q., Ren, T.B., Song, A.D., Zuo, R.Y., Guo, H.W., 2012b. Effect of steam explosion pretreatment and microbial fermentation on degradation of corn straw. *Advanced Materials Research* 343, 809–814.
- Chen, W., Zhang, S., Li, Y., Wu, H., Meng, Q., Zhou, Z., 2019. Steam-exploded sugarcane bagasse as a potential beef cattle feedstock: effects of different pretreatment conditions. *Journal of Animal Science* 97, 2414–2423. <https://doi.org/10.1093/jas/skz127>.
- Du, C., Nan, X., Wang, K., Zhao, Y., Xiong, B., 2019. Evaluation of the digestibility of steam-exploded wheat straw by ruminal fermentation, sugar yield and microbial structure *in vitro*. *RSC Advances* 9, 41775–41782. <https://doi.org/10.1039/c9ra08167d>.
- Fernandez, C., Romero, T., Marti, J.V., Moya, V.J., Hernando, I., Llor, J.J., 2021. Energy, nitrogen partitioning, and methane emissions in dairy goats differ when an isoenergetic and isoproteic diet contained orange leaves and rice straw crop residues. *Journal of Dairy Science* 104, 7830–7844. <https://doi.org/10.3168/jds.2020-19953>.
- Fjortoft, K., Morken, J., Hanssen, J.F., Briseid, T., 2019. Pre-treatment methods for farm-scale biogas plants. *Biomass and Bioenergy* 124, 88–94.
- Getahun, A., Kechero, Y., Yemane, N., Dessie, T., Esatu, W., 2025. Nutritional evaluation and potential of locally available alternative feed resources for sustainable poultry production: a case study of smallholder farms in central and southern Ethiopia. *Tropical Animal Health and Production* 57, 317. <https://doi.org/10.1007/s11250-025-04549-7>.
- Grabber, J.H., 2005. How do lignin composition, structure, and cross-linking affect degradability? A review of cell wall model studies. *Crop Science* 45, 820–831.
- Greenwood, P.L., 2021. Review: an overview of beef production from pasture and feedlot globally, as demand for beef and the need for sustainable practices increase. *Animal* 15, 100295. <https://doi.org/10.1016/j.animal.2021.100295>.
- He, L., Wu, H., Wang, G., Meng, Q., Zhou, Z., 2018. The effects of including corn silage, corn stalk silage, and corn grain in finishing ration of beef steers on meat quality and oxidative stability. *Meat Science* 139, 142–148. <https://doi.org/10.1016/j.meatsci.2018.01.023>.
- He, L., Huang, Y., Shi, L., Zhou, Z., Wu, H., 2022. Steam explosion processing intensifies the nutritional values of most crop byproducts: morphological structure, carbohydrate-protein fractions, and rumen fermentation profile. *Frontiers in Nutrition* 9, 979609. <https://doi.org/10.3389/fnut.2022.979609>.
- Hussein, A.M., Hassanien, H., Phillip, Y.L., Abou, E.M., El-Badawy, M.M., El-Maghraby, M.M., Khayyal, A.A., Salem, A., 2024. Effect of urea-treated rice straw, mixed with faba bean straw, on nutrient digestibility, blood metabolites and performance of growing lambs. *Tropical Animal Health and Production* 56, 122. <https://doi.org/10.1007/s11250-024-03971-7>.
- Jacquet, N., Maniet, G., Vanderghem, C., Delvigne, F., Richel, A., 2015. Application of steam explosion as pretreatment on lignocellulosic material: a review. *Industrial & Engineering Chemistry Research* 54, 2593–2598.
- Li, H., Dai, M., Dai, S., Dong, X., 2018. Current status and environment impact of direct straw return in China's cropland - a review. *Ecotoxicology and Environmental Safety* 159, 293–300. <https://doi.org/10.1016/j.ecoenv.2018.05.014>.
- Li, J., Xi, H., Wang, A., Nie, M., Gong, X., Lin, R., Zhang, X., Tian, Y., Wang, F., Tong, L.T., 2024. Effects of high-pressure microfluidization treatment on the structural, physicochemical properties of insoluble dietary fiber in highland barley bran. *International Journal of Biological Macromolecules* 262, 129743. <https://doi.org/10.1016/j.ijbiomac.2024.129743>.
- Li, J., Wang, H., Liang, R., Mao, Y., Yang, X., Luo, X., Zhu, L., Zhang, Y., 2025. Effect of super-fast freezing before rigor mortis on quality of bovine m. Longissimus Lumborum. *Meat Science* 228, 109880. <https://doi.org/10.1016/j.meatsci.2025.109880>.
- Llor, J.J., Elolimy, A.A., McCann, J.C., 2016. Dietary impacts on rumen microbiota in beef and dairy production. *Animal Frontiers* 6, 22–29.
- National Academies of Sciences, Engineering, and Medicine (NASEM), 2016. *Nutrient Requirements of Beef Cattle*. 8th ed. Animal Nutrition Series. National Academy Press, Washington, DC, USA.
- Nie, D., Yao, L., Xu, X., Zhang, Z., Li, Y., 2021. Promoting corn stover degradation via sequential processing of steam explosion and cellulase/lactic acid bacteria-assisted ensilage. *Bioresource Technology* 337, 125392. <https://doi.org/10.1016/j.biortech.2021.125392>.
- Ørskov, E., McDonald, I., 1979. The estimation of protein degradability in the rumen from incubation measurements weighted according to rate of passage. *The Journal of Agricultural Science* 92, 499–503.
- Pan, S., Wang, D., Lin, Y., Cheng, M., Zhu, F., Guo, Y., 2024. Effects of ginger straw silage with enzymes on growth performance, digestion and metabolism, meat quality and rumen microflora diversity of laiwu black goat. *Animals* 14, 2040. <https://doi.org/10.3390/ani14142040>.
- Rosenson, R.S., Brewer, H.J., Davidson, W.S., Fayad, Z.A., Fuster, V., Goldstein, J., Hellerstein, M., Jiang, X.C., Phillips, M.C., Rader, D.J., Remaley, A.T., Rothblat, G.H., Tall, A.R., Yvan-Charvet, L., 2012. Cholesterol efflux and atheroprotection: advancing the concept of reverse cholesterol transport. *Circulation* 125, 1905–1919. <https://doi.org/10.1161/CIRCULATIONAHA.111.066589>.
- Rothmann, C., Rothmann, L., Viljoen, B., Cason, E.D., 2023. Application of solid-state fermentation using mushrooms for the production of animal feed. *Journal of Basic Microbiology* 63, 1153–1164. <https://doi.org/10.1002/jobm.202300218>.
- Shi, Q., Li, Y., Li, Y., Cheng, Y., Zhu, W., 2019. Effects of steam explosion on lignocellulosic degradation of, and methane production from, corn stover by a co-cultured anaerobic fungus and methanogen. *Bioresource Technology* 290, 121796. <https://doi.org/10.1016/j.biortech.2019.121796>.
- Van Soest, P.J., Robertson, J.B., 1981. The detergent system of analysis and its application to human foods. In: James, W.P.T., Theander, O. (Eds.), *The Analysis of Dietary Fiber in Foods*. Marcel Dekker, New York, NY, USA, pp. 123–158.
- Van Soest, P.J., Robertson, J.B., Lewis, B.A., 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *Journal of Dairy Science* 74, 3583–3597. [https://doi.org/10.3168/jds.S0022-0302\(91\)78551-2](https://doi.org/10.3168/jds.S0022-0302(91)78551-2).
- Wang, F., Li, D.Y., Meng, Q.X., 2016. Technical note: a special apparatus for facilitating the in situ nylon bag measurement of the ruminal degradation of feedstuffs in cattle. *Journal of Animal Science* 94, 3457–3463. <https://doi.org/10.2527/jas.2016-0574>.
- Wang, Y., Yin, X., Hao, K., Wang, C., Wei, W., Li, Y., Gao, S., Ji, Z., Wang, W., Xie, Y., Li, C., 2025. Consumption of steam explosion and fermentation-pretreated corn stover affects the growth performance of sheep by shifting the rumen microbiota community structure. *Frontiers in Microbiology* 16, 1532746. <https://doi.org/10.3389/fmicb.2025.1532746>.
- Xu, M., Chen, X., Huang, Z., Chen, D., Li, M., He, J., Chen, H., Zheng, P., Yu, J., Luo, Y., Yu, B., 2022. Effects of dietary grape seed proanthocyanidin extract supplementation on meat quality, muscle fiber characteristics and antioxidant capacity of finishing pigs. *Food Chemistry* 367, 130781. <https://doi.org/10.1016/j.foodchem.2021.130781>.
- Yang, D., Sun, T., Zheng, Y., Ghaffari, M.H., Huang, Y., Sun, Y., Li, X., Ma, T., 2025. Effect of replacing peanut vine with extruded rape straw on growth, nutrient digestibility, energy metabolism, microbial crude protein synthesis, meat amino acid and fatty acid profiles of finishing lambs. *Translational Animal Science* 9, txa044. <https://doi.org/10.1093/tas/txa044>.
- Yang, X., Zhang, Y., Luo, X., Zhang, Y., Zhu, L., Xu, B., Hopkins, D.L., Liang, R., 2022. Influence of oxygen concentration on the fresh and internal cooked color of modified atmosphere packaged dark-cutting beef stored under chilled and superchilled conditions. *Meat Science* 188, 108773. <https://doi.org/10.1016/j.meatsci.2022.108773>.
- Yi, S., Hu, S., Wang, J., Abudukelimu, A., Wang, Y., Li, X., Wu, H., Meng, Q., Zhou, Z., 2024a. Effect of guanidinoacetic acid supplementation on growth performance, rumen fermentation, blood indices, nutrient digestion, and nitrogen metabolism in angus steers. *Animals* 14, 401. <https://doi.org/10.3390/ani14030401>.
- Yi, S., Ye, B., Wang, J., Yi, X., Wang, Y., Abudukelimu, A., Wu, H., Meng, Q., Zhou, Z., 2024b. Investigation of guanidinoacetic acid and rumen-protected methionine induced improvements in longissimus lumborum muscle quality in beef cattle. *Meat Science* 217, 109624. <https://doi.org/10.1016/j.meatsci.2024b.109624>.
- Yi, S., Wang, J., Ye, B., Yi, X., Abudukelimu, A., Wu, H., Meng, Q., Zhou, Z., 2025. Guanidinoacetic acid and methionine supplementation improve the growth performance of beef cattle via regulating the antioxidant levels and protein and lipid metabolisms in serum and liver. *Antioxidants* 14, 559. <https://doi.org/10.3390/antiox14050559>.
- Yu, H., Chen, X., He, J., Zheng, P., Luo, Y., Yu, B., Chen, D., Huang, Z., 2025. Effects of dietary grape seed proanthocyanidin extract supplementation on muscle water-holding capacity of finishing pigs. *Meat Science* 227, 109865. <https://doi.org/10.1016/j.meatsci.2025.109865>.
- Zhang, Y., Holman, B., Mao, Y., Chen, X., Luo, X., Hopkins, D.L., Zhang, Y., 2021. Determination of a pH threshold for dark cutting beef based on visual evaluation by Asian consumers. *Meat Science* 172, 108347. <https://doi.org/10.1016/j.meatsci.2020.108347>.
- Zhang, T., Jiang, D., Li, Y., Zhang, H., Zhang, Z., Jing, Y., Lu, C., Zhang, Y., Xia, C., Zhang, Q., 2022. Lignin removal, reducing sugar yield and photo-fermentative biohydrogen production capability of corn stover: effects of different pretreatments. *Bioresource Technology* 346, 126437. <https://doi.org/10.1016/j.biortech.2021.126437>.
- Zhang, L., Liu, Y., Hao, L., 2016. Contributions of open crop straw burning emissions to pm2.5 concentrations in China. *Environmental Research Letters* 11, 14014.
- Zhang, H., Zhang, R., Song, Y., Miu, X., Zhang, Q., Qu, J., Sun, Y., 2023. Enhanced enzymatic saccharification and ethanol production of corn stover via pretreatment with urea and steam explosion. *Bioresource Technology* 376, 128856. <https://doi.org/10.1016/j.biortech.2023.128856>.
- Zhao, S., Li, G., Zheng, N., Wang, J., Yu, Z., 2018. Steam explosion enhances digestibility and fermentation of corn stover by facilitating ruminal microbial colonization. *Bioresource Technology* 253, 244–251. <https://doi.org/10.1016/j.biortech.2018.01.024>.
- Zhao, X., Wang, F., Fang, Y., Zhou, D., Wang, S., Wu, D., Wang, L., Zhong, R., 2020. High-potency white-rot fungal strains and duration of fermentation to optimize corn straw as ruminant feed. *Bioresource Technology* 312, 123512. <https://doi.org/10.1016/j.biortech.2020.123512>.